AM Hacks

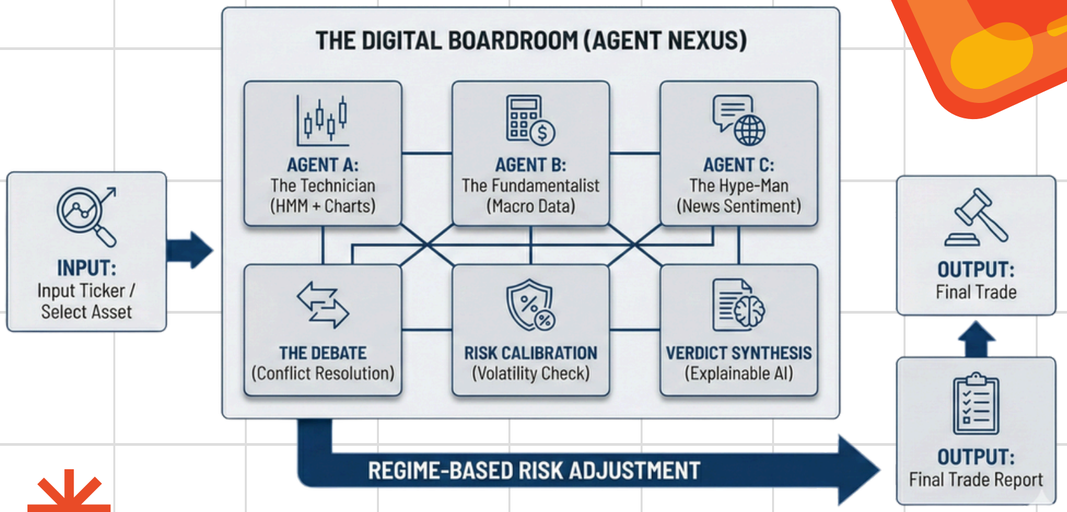
- **Theme:** FinTech
- **Problem Statement:** Solving the "Regime Blindness" and "Black Box" crisis in Financial AI.
- **Track:** Open Innovation and TRAE
- **Team Name:** Anomaly
- **Project Name:** QuantPulse
- **Team Members:**
  - Riddhi Sharma(TeamLeader)
  - Raja Rathour

QuantPulse

PROBLEM

India's retail investment landscape is currently facing a "Crisis of Speculation," where over 91% of individual traders in the F&O segment incur losses, wiping out lakhs of crores in national wealth annually. The Union Budget 2026 has explicitly targeted this issue, implementing sharp hikes in Securities Transaction Tax (STT) to disincentivize reckless gambling and push the market toward maturity and "wiseful trading." However, a critical gap remains: while current brokerage platforms have successfully democratized access to the markets, they have completely failed to democratize wisdom. Retail investors are currently left drowning in conflicting news, raw data, and noise, forcing them to trade on impulse rather than probability. There is simply no existing tool that acts as a "Personal Risk Manager" for the common man. QuantPulse bridges this specific gap by replacing emotional speculation with an "Agentic AI" ensemble, directly aligning with the government's mandate for financial stability by filtering noise and enforcing mathematical rigor before a trade is ever placed.

USER FLOW



PROPOSED SOLUTION

Regime-Adaptive Forecasting

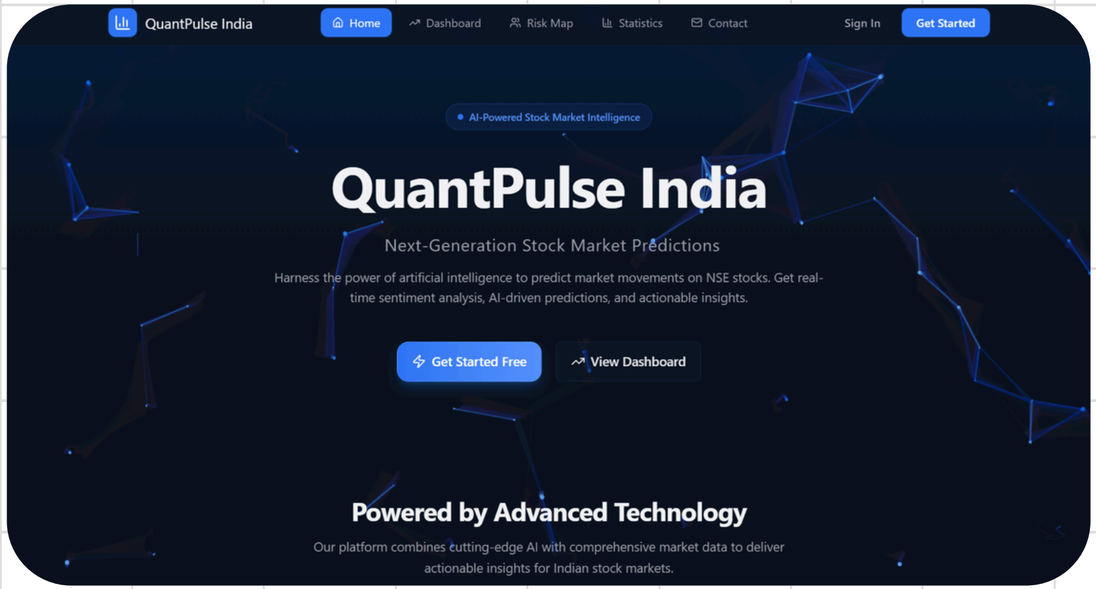
Weuse UnsupervisedLearning **(HMMs)** tofirstdiagnosethe 'Market Weather' (Panic vs. Growth), ensuring the **model adapts** its risk parameters dynamically.

The Digital Boardroom

We replace the single model with a **multi-agent team** (A Chartist, A Fundamentalist, and A Skeptic) that debates the trade. A final 'Judge' synthesizes their arguments into an explainable verdict.

Glass-Box" AI

Unlike black-boxLSTMs, oursystem provides a written **'Verdict'** explaining exactly why a trade wasrecommended (e.g.,'Technicalsare bullish,but Macro risks are too high').

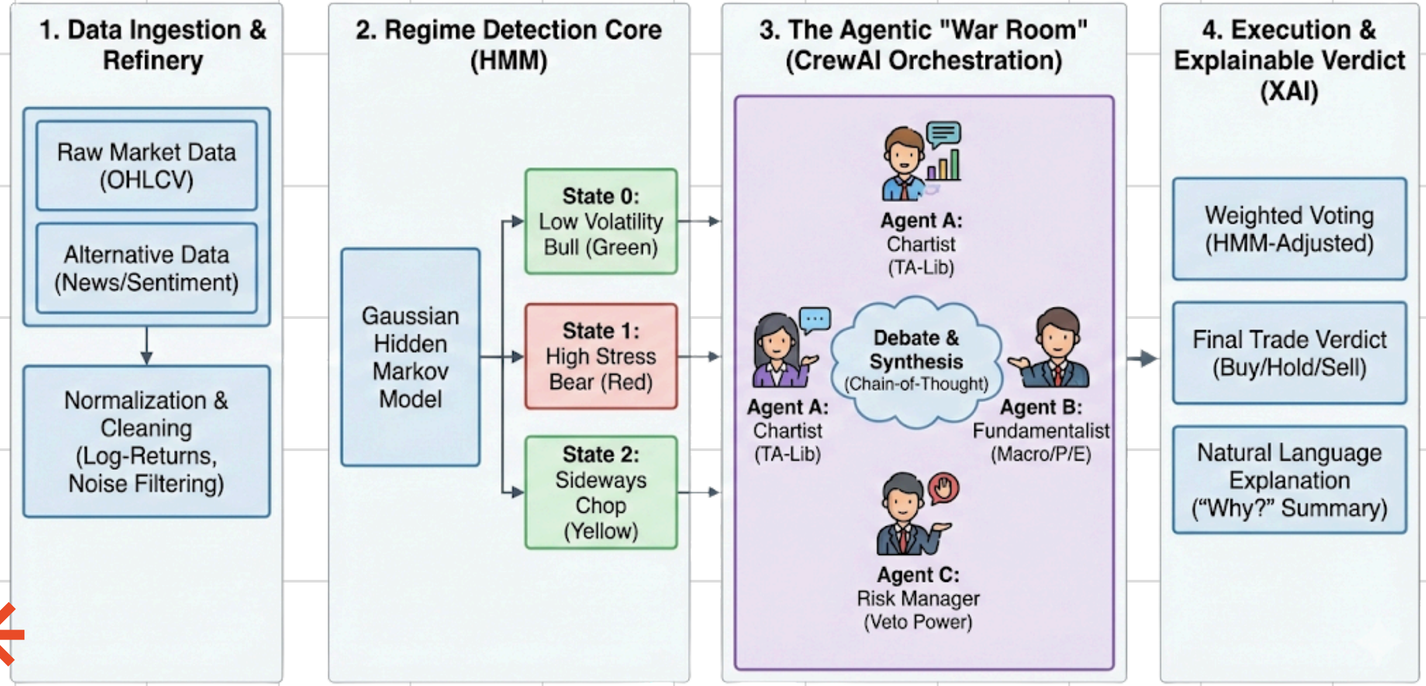


Technical Approach

PIPELINE

QuantPulse's architecture transforms raw market noise into actionable wisdom through a sophisticated four-stage pipeline. It begins with Data Ingestion, where multi-source inputs (OHLCV prices and alternative news data) undergo rigorous normalization and noise filtering to ensure statistical purity. This refined data feeds into a Context-Aware Embedding Layer, which utilizes a Gaussian Hidden Markov Model (HMM) to detect hidden market regimes (such as "High Volatility" or "Bull Run"), creating a dynamic backdrop for decision-making. The core innovation is the Reasoning Engine ("The War Room"), where a CrewAI-orchestrated ensemble of specialized agents - a Technician, a Fundamentalist, and a risk-focused Skeptic - engage in a chain-of-thought debate to synthesize conflicting signals. Finally, the pipeline delivers a transparent Output Layer, presenting the user with a weighted trade verdict, a natural language "Why?" explanation, and a risk confidence score via the interactive dashboard.

QuantPulse: Regime-Adaptive Agentic Pipelin



TECH STACK

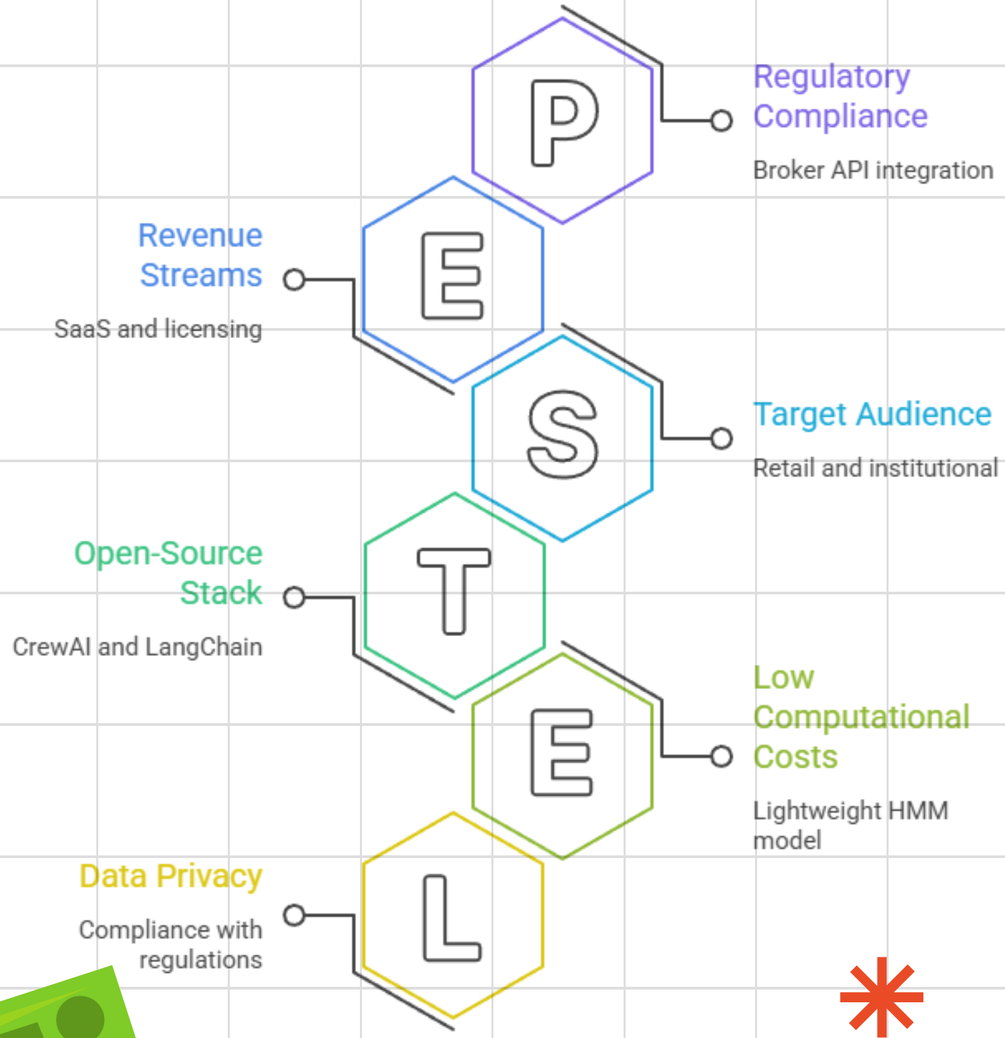
crewai, python, PyTorch, pandas, FastAPI, JS, CSS, HTML, LangChain, matplotlib, scikit learn

NOVELTY

- We solve the **"Black Box" problem** by replacing opaque predictions with a transparent **Agentic Debate**, fusing Quantitative rigor with Qualitative reasoning into a single output. Eliminates **"Regime Blindness"** by using HMMs to dynamically adjust agent behavior based on the current market state (Panic vs. Growth), preventing "Bull Market models" from failing in crashes. Our approach is **interpretable by design**, delivering a written "Investment Memo" alongside the prediction, mimicking the workflow of a real-world Hedge Fund Committee.

Features And Viability

AI Trading Agent Platform PESTEL Analysis



Future Scope

Live Execution: Phase 2 will connect the 'Judge Agent' directly to Broker APIs (e.g., Zerodha, Alpaca) to execute trades automatically, not just recommend them.

Key Challenges in Data Processing

API Rate Limits

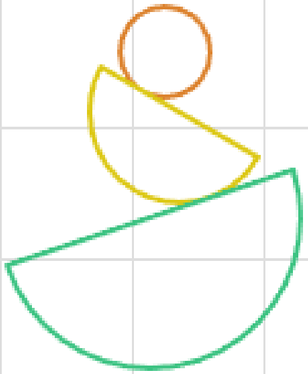
Restricts data retrieval speed

Incomplete Databases

Skews analysis results

Validating Novelty

Requires expert review



Feasibility at a Glance

1. High Interpretability
2. Solves "Trust Gap"
3. Low API/Compute Cost
4. Human-Like Reasoning
5. Crash-Resistant Logic

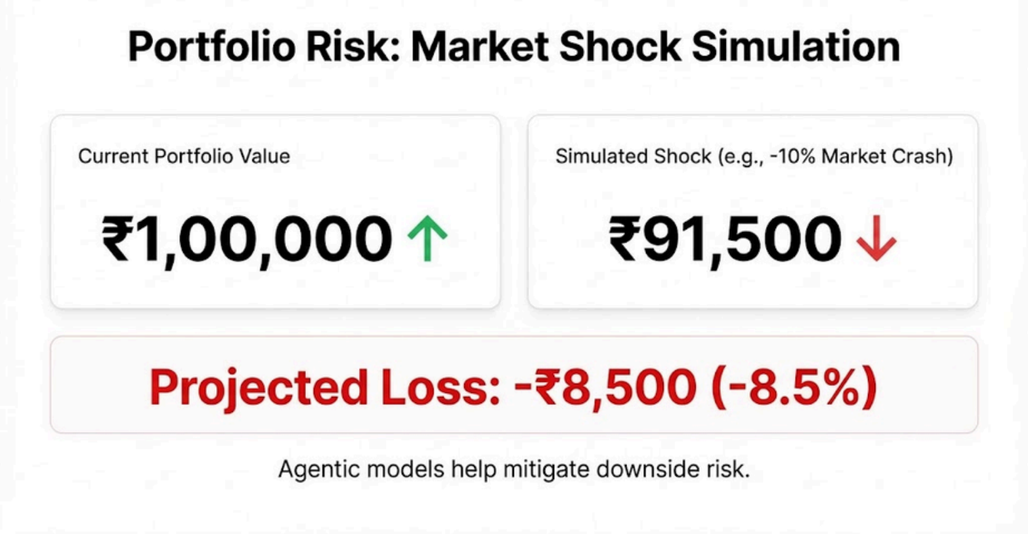
USP and Competitor Analysis

Demo

Website

GitHub

SRSD



USP & Innovation: The "Agentic AI" Advantage

- Multi-Agent Ensemble: Unlike simple predictors, we use a committee of AI agents (Technical, News, and Market Topology) to debate and weight predictions.
- Explainable AI (XAI): We don't just give a number. We show why the prediction was made by displaying the weighted influence of each agent (e.g., "News Sentiment adjusted the base forecast by +1.5%").
- Institutional-Grade Risk Tools: Features like "Simulate Market Shock" allow users to stress-test portfolios against hypothetical crashes, moving beyond simple tracking.

Competitor Analysis: Why We Are Better

- vs. Traditional Platforms (Yahoo Finance): They are cluttered and backward-looking. QuantPulse is a clean, modern dashboard focused on forward-looking, actionable insights.
- vs. Professional Tools (TradingView): They are powerful but complex with a steep learning curve. QuantPulse delivers pre-packaged, high-level AI insights without requiring custom scripting.

Research And References

1. The Core Problem: "91% Loss" Crisis

Use these links to prove the "Financial Health Crisis" among retail traders.

SEBI FY25 Report (91% Loss Statistic):

Fact: 91% of individual traders in the Equity Futures & Options (F&O) segment incurred net losses in FY 2024-25.

Fact: The total loss incurred by these retail traders was ₹1.06 Lakh Crore, a massive drain on household savings.

Fact: Despite losing money, over 75% of these loss-making traders continued to trade, proving the "addictive/gambling" nature of the current market.

Reference Link: SEBI Study on Retail F&O Losses (FY25)

2. The Regulatory Trigger: Budget 2026 & STT Hike

Use these links to argue that the government is actively punishing speculation and demanding "Wiseful" tools.

Union Budget 2026 (Securities Transaction Tax Hike):

Fact: The Finance Minister hiked STT on Futures by 150% (from 0.02% to 0.05%) and on Options by 50% (from 0.1% to 0.15%).

Fact: The stated goal of this policy is to "curb excessive speculation" and discourage "retail gambling" in the derivatives market.

Reference Link: Budget 2026: STT Hike on F&O Explained

Reference Link: FM Sitharaman: STT Hike Meant to Protect Retail Investors

3. The "Gamification" Crackdown

Use these to show that competitors who "gamify" trading are being banned, leaving a gap for your "Serious/Mathematical" tool.

SEBI vs. Fantasy/Virtual Trading:

Fact: SEBI has issued warnings and stopped data sharing with "Virtual Trading" and "Fantasy Stock" apps because they encourage reckless behavior without real risk management.

Reference Link: SEBI Advisory on Unauthorised Virtual Gaming Platforms

4. "Wiseful Trading" & Risk Management (The Solution Gap)

Use these to support your "Agentic AI" solution.

Need for Institutional Tools for Retail:

Fact: Institutional investors (foreign investors, props) use algorithms and hedging to make money, while retail traders (who lack these tools) bear the losses.

Fact: SEBI's new risk management rules for 2025 emphasize "Delta-based Open Interest" and stricter position monitoring—complex metrics that your tool can simplify for users.

Reference Link: SEBI Risk Management Rules for Derivatives 2025